

IEM Master Graduation Assignment at ING

Robust and Stable Evidence Retrieval in Retrieval-Augmented Generation

Background and Motivation

Retrieval-Augmented Generation (RAG) systems are designed to generate answers supported by verifiable evidence. However, conventional RAG pipelines can be sensitive to repeated runs of the same question or to small, meaning-preserving variations in how a question is phrased. Due to the non-deterministic behavior of LLM-based components, semantically equivalent questions may retrieve different documents or pages, produce different citations, and ultimately lead to answers with different wording or even different conclusions.

At ING, this challenge is particularly relevant in applications such as Know Your Customer (KYC), where AI-generated answers need to be traceable, reproducible, and auditable. Users should be able to expect that repeated or semantically equivalent questions produce consistent conclusions supported by the same, or equally good, evidence. At a minimum, any remaining variability should be measurable and explainable. Improving the stability of RAG systems can therefore contribute to more reliable and auditable AI-assisted workflows in financial applications.

Approach

This project investigates how to measure and improve the stability of a RAG pipeline under repeated and semantically equivalent queries. The research will consider stability at three interconnected levels:

- Retrieval stability: Semantically equivalent versions of a question should lead to the same relevant sources being provided to the generator (LLM).
- Evidence stability: The final citations selected to support the answer should refer to the same source passages or pages and substantiate the same factual claims.
- Answer stability: The answers, reasoning, and conclusions derived from the evidence should remain equivalent in meaning and, when output requirements are fixed, reasonably consistent in completeness, length, and style.

The study will define stability metrics, construct sets of questions for repeated runs and meaning-preserving perturbations, identify the main sources of instability in a RAG pipeline, and experimentally compare mitigation strategies. The intended outcome is an evaluated approach together with practical recommendations for building more reproducible and auditable RAG systems.

Research Questions

How can the stability of retrieval-augmented generation systems be measured and improved under repeated and semantically equivalent queries?

- RQ1: How can retrieval, evidence, and answer stability in RAG systems be measured?

- RQ2: What are the main sources of instability across repeated runs and meaning-preserving query variations?
- RQ3: Which mitigation strategies are most effective at improving RAG stability?
- RQ4: What trade-offs arise between stability and the quality of retrieved evidence and generated answers?

Student Profile

We are looking for a motivated MSc student with:

- Solid Python programming skills.
- Experience with data analysis and experimentation.
- Completed courses in Machine Learning, Data Mining, Artificial Intelligence, Statistics, or a related field.
- Interest in LLMs, RAG systems, and trustworthy AI.
- Interest in applying AI research to real-world business problems and willingness to collaborate with researchers and practitioners from ING.

Planning

Weeks 1–4 Literature Review and Problem Formulation

Weeks 5–8 Stability Metrics and Experimental Setup

Weeks 9–14 Baselines and Instability Analysis

Weeks 15–20 Mitigation Strategies and Evaluation

Weeks 21–24 Analysis and Practical Recommendations

Weeks 25–30 Writing

Practical Information

- Organization: University of Twente, in collaboration with ING.
- Working arrangement: Hybrid, with approximately one day per week at ING's Amsterdam office.
- Internship: The student will join ING as an MSc thesis intern and receive an **internship allowance**.
- Duration: 30 weeks.
- Supervision: Joint supervision by researchers from ING and the University of Twente.
- Start date: Preferably as soon as possible, with flexibility depending on the student's availability.
- Contact: Xutong Liu, Postdoctoral Researcher, University of Twente, xutong.liu@utwente.nl